MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE RHG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic inverter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.049 failures per million operating hours for class I or 0.121 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RHG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/298-1 for class I and M28787/298-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

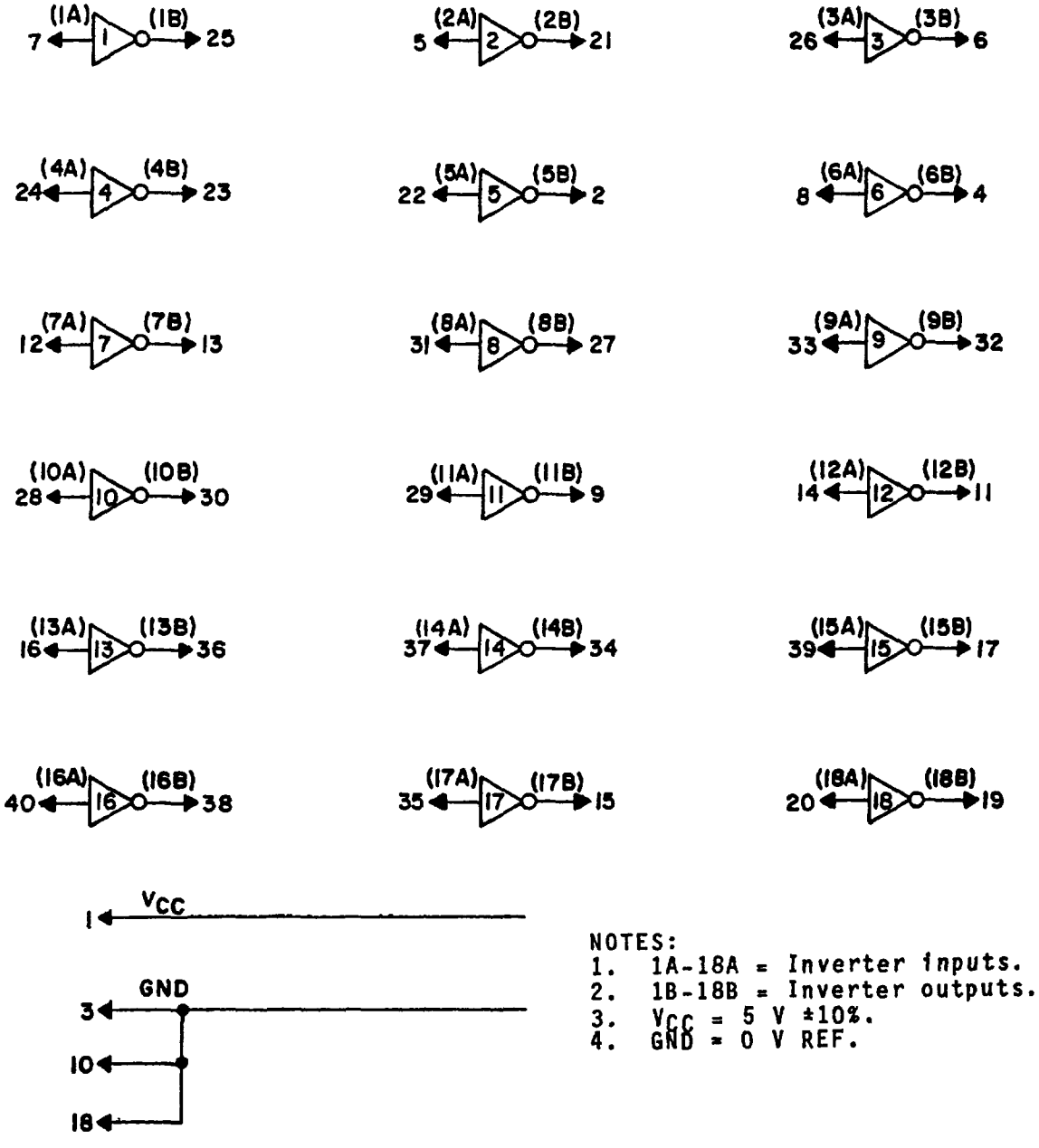
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-18A = Inverter inputs.
2. 1B-18B = Inverter outputs.
3. V_{CC} = 5 V ±10%.
4. GND = 0 V REF.

FIGURE 1. RHG module function diagram, inverter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B18	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C18	Input breakdown current	V	S	0	100	0	100	0	100	0	100	0	100	µA
D1-D18	Output leakage current	VI	X	0	1	0	1	0	1	0	1	0	1	mA
E1-E18	Output short circuit current	VII	X	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
F1-F18	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G18	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H18	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J18	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
K1-K18	Input current, low level	XII	X	-10	-430	-10	-430	-10	-430	-10	-430	-10	-430	µA
L1,L2	Ground pin continuity	XIII	X	0	1	0	1	0	1	0	1	0	1	Ω
L3,L4	Ground pin continuity	XIII	Q	0.00	0.10	0.00	0.10	0.00	0.10	0.00	0.10	0.00	0.10	Ω
M1	Power supply current, low level	XIV	S	0.1	19.8	0.1	19.8	0.1	19.8	0.1	19.8	0.1	19.8	mA
M2	Power supply current, high level	XIV	S	0.1	7.2	0.1	7.2	0.1	7.2	0.1	7.2	0.1	7.2	mA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF	XV	S	5	50	5	50	5	55	5	50	5	60	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	10	80	10	80	10	90	10	80	10	100	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF	XV	S	3	18	3	18	3	20	3	18	3	22	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P19-P36	Transition time (t_{THL}), C_L = 150 pF	XV	Q	6	40	6	40	6	42	6	40	6	44	ns
Q1-Q18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	3	19	3	19	3	21	3	19	3	23	ns
Q19-Q36	Propagation delay time (t_{PLH}), C_L = 150 pF	XVI	Q	4	29	4	29	4	31	4	29	4	33	ns
R1-R18	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	3	19	3	19	3	21	3	19	3	23	ns
R19-R36	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	4	29	4	29	4	31	4	29	4	33	ns
S1-S18	Output crosstalk, low level	XVII	Q	0	650	0	650	0	650	0	650	0	650	mV
S19-S36	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
T1-T18	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XVIII	Q	0	500	0	500	0	500	0	500	0	500	mV
V	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W18	Input capacitance	XX	Q	5	15	5	15	5	15	5	15	5	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T and V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation			
All outputs low (outputs not loaded)	109	109	mW
All outputs high (outputs not loaded)	40	40	mW
Supply voltage	5 V dc ±10%	5 V dc ±10%	

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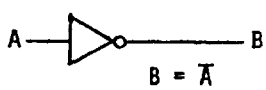
	Class I	Class II	Unit
Input voltage			
Low level	-500 mV min, 700 mV max	-500 mV min, 700 mV max	
High level	2 V min, V_{CC} max	2 V min, V_{CC} max	
Input current			
Low level (V_{IL} = 0 V)	-430 max	-430 max	mA
High level (V_{IH} = 2.7 V)	20 max	20 max	µA
Output voltage			
Low level (I_{OL} = 4 mA)	400 max	400 max	mV
High level (I_{OH} = -400 µA)	2.5 min	2.5 min	V
Output drive current			
Low level (V_{OL} = 400 mV)	4 max	4 max	mA
High level (V_{OH} = 2.5 V)	-400 max	-400 max	µA

6.5 Module function. This module contains 18 independent logic inverters.

6.5.1 Input to output relationship. Each inverter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input A	Output B
L	H
H	L



6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. Circuits 1,2,3,4,5, and 6.
b. Circuits 7,8,9,10,11 and 12.
c. Circuits 13,14,15,16,17, and 18.

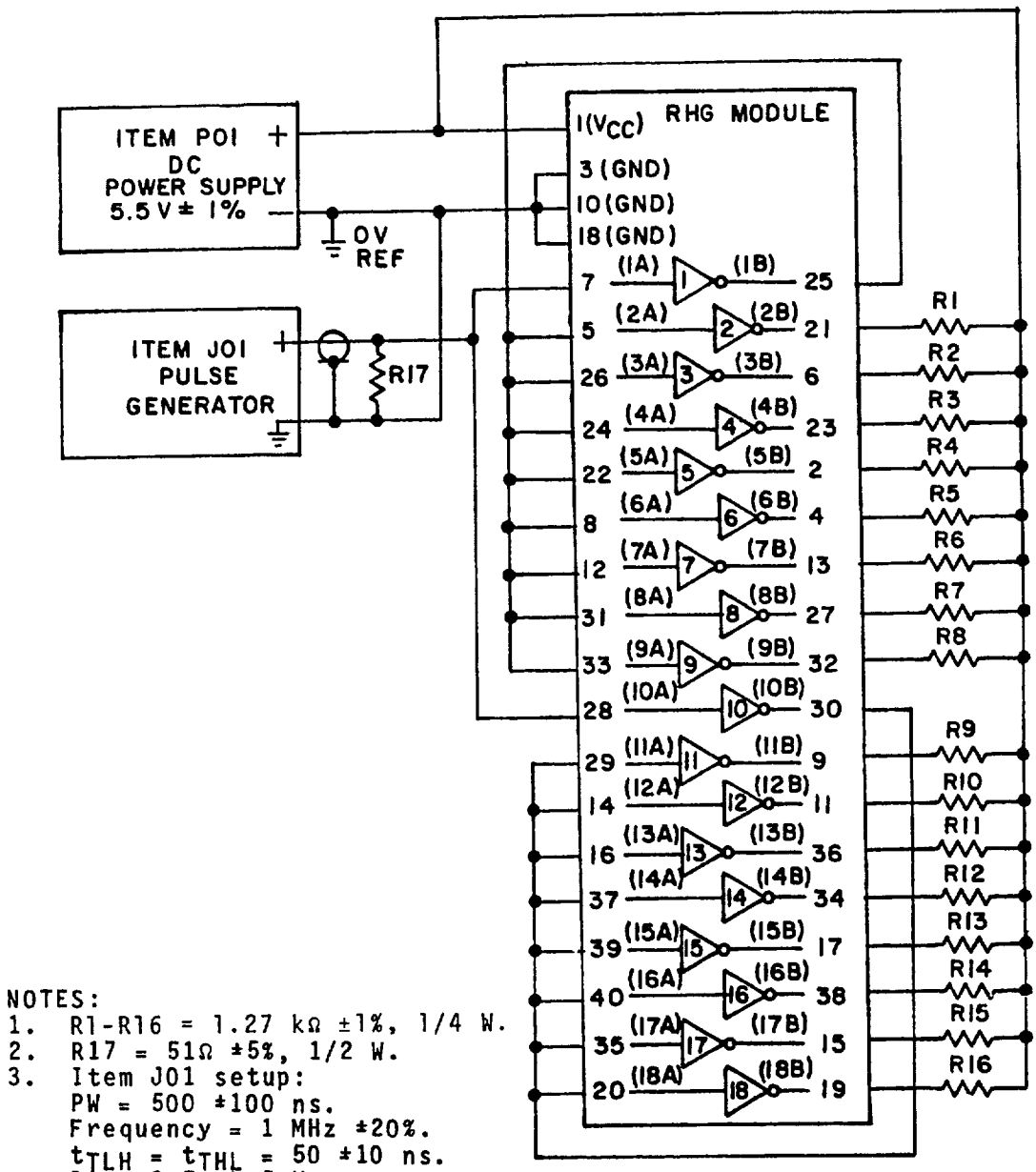
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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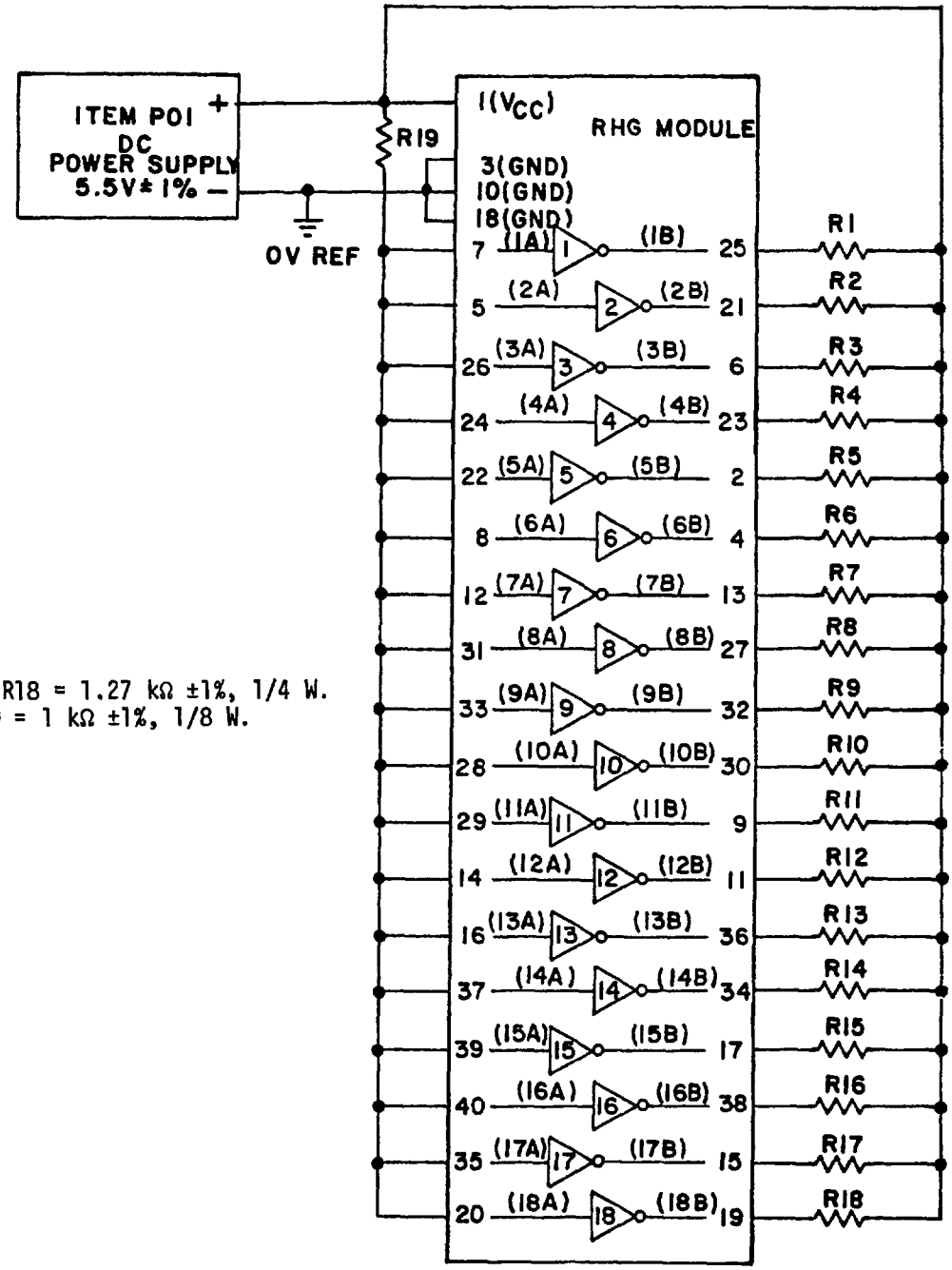
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NOTES:
1. R1-R16 = 1.27 kΩ ±1%, 1/4 W.
2. R17 = 51Ω ±5%, 1/2 W.
3. Item J01 setup:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R18 = 1.27 kΩ ±1%, 1/4 W.
2. R19 = 1 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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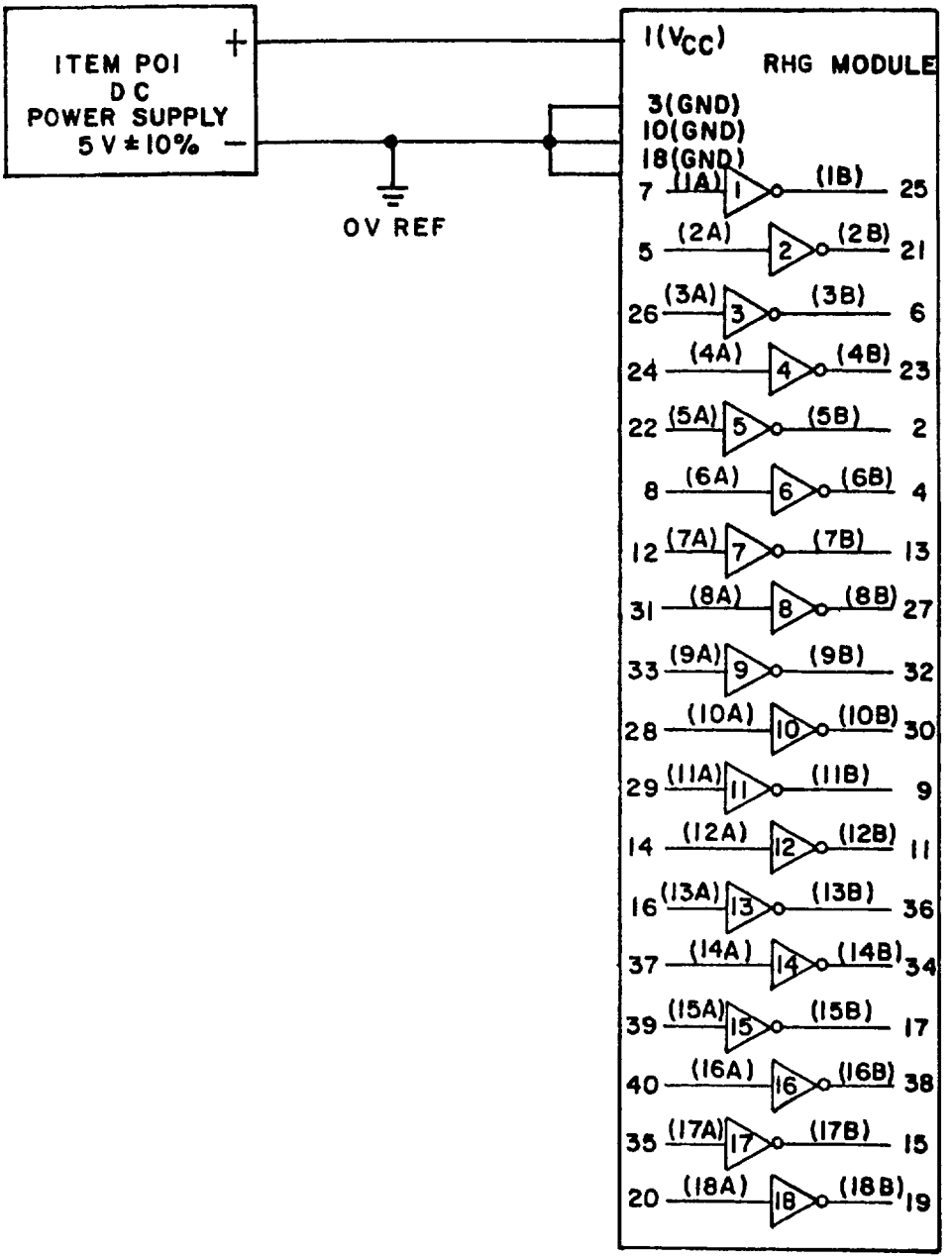


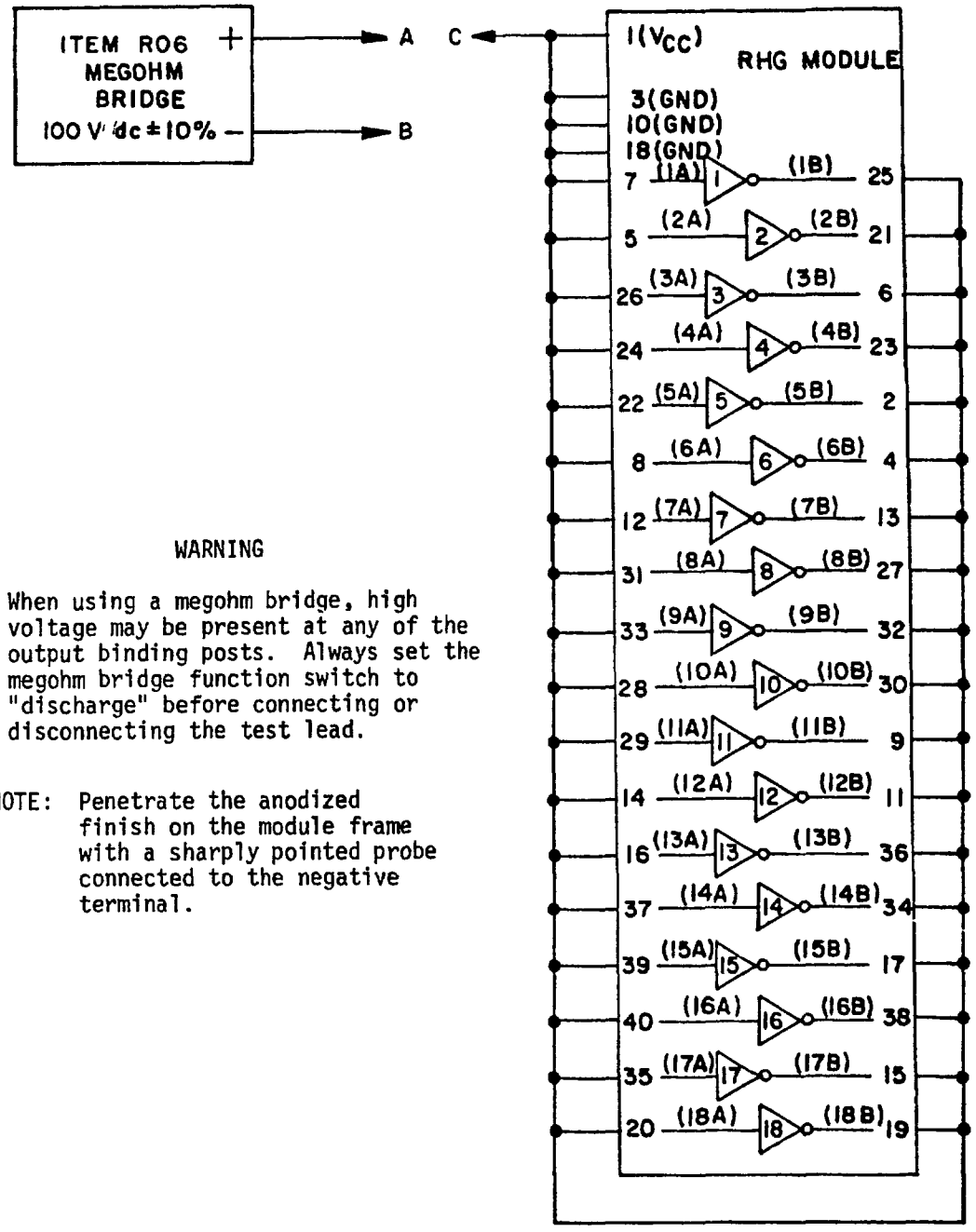
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test lead.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

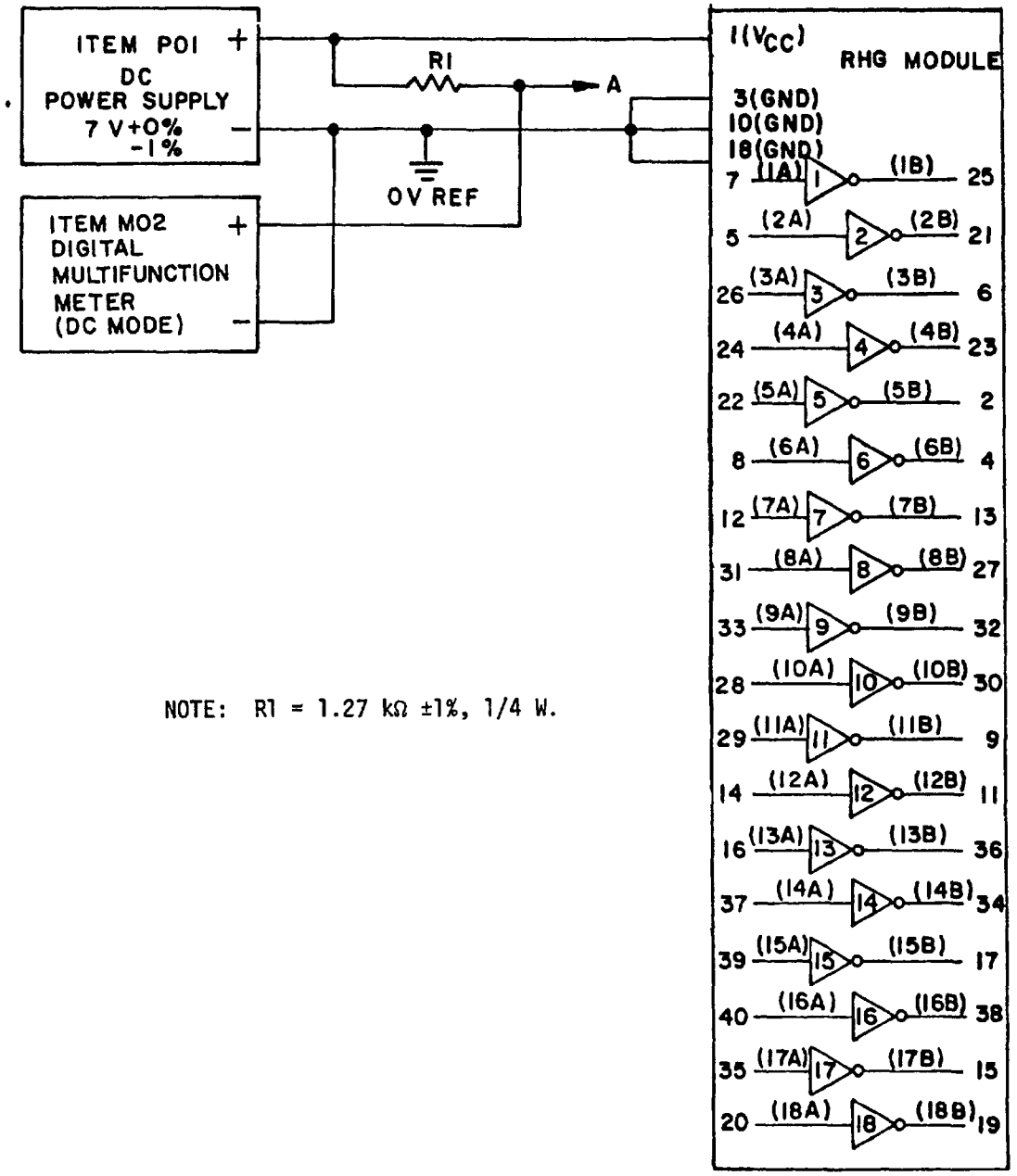
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
B1	1B V_{OUT} (low loaded)	25	10	800	mV	6
B2	2B "	21	"	"	"	"
B3	3B "	6	"	"	"	"
B4	4B "	23	"	"	"	"
B5	5B "	2	"	"	"	"
B6	6B "	4	"	"	"	"
B7	7B "	13	"	"	"	"
B8	8B "	27	"	"	"	"
B9	9B "	32	"	"	"	"
B10	10B "	30	"	"	"	"
B11	11B "	9	"	"	"	"
B12	12B "	11	"	"	"	"
B13	13B "	36	"	"	"	"
B14	14B "	34	"	"	"	"
B15	15B "	17	"	"	"	"
B16	16B "	38	"	"	"	"
B17	17B "	15	"	"	"	"
B18	18B "	19	"	"	"	"

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NOTE: R1 = 1.27 kΩ ±1%, 1/4 W.

*FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

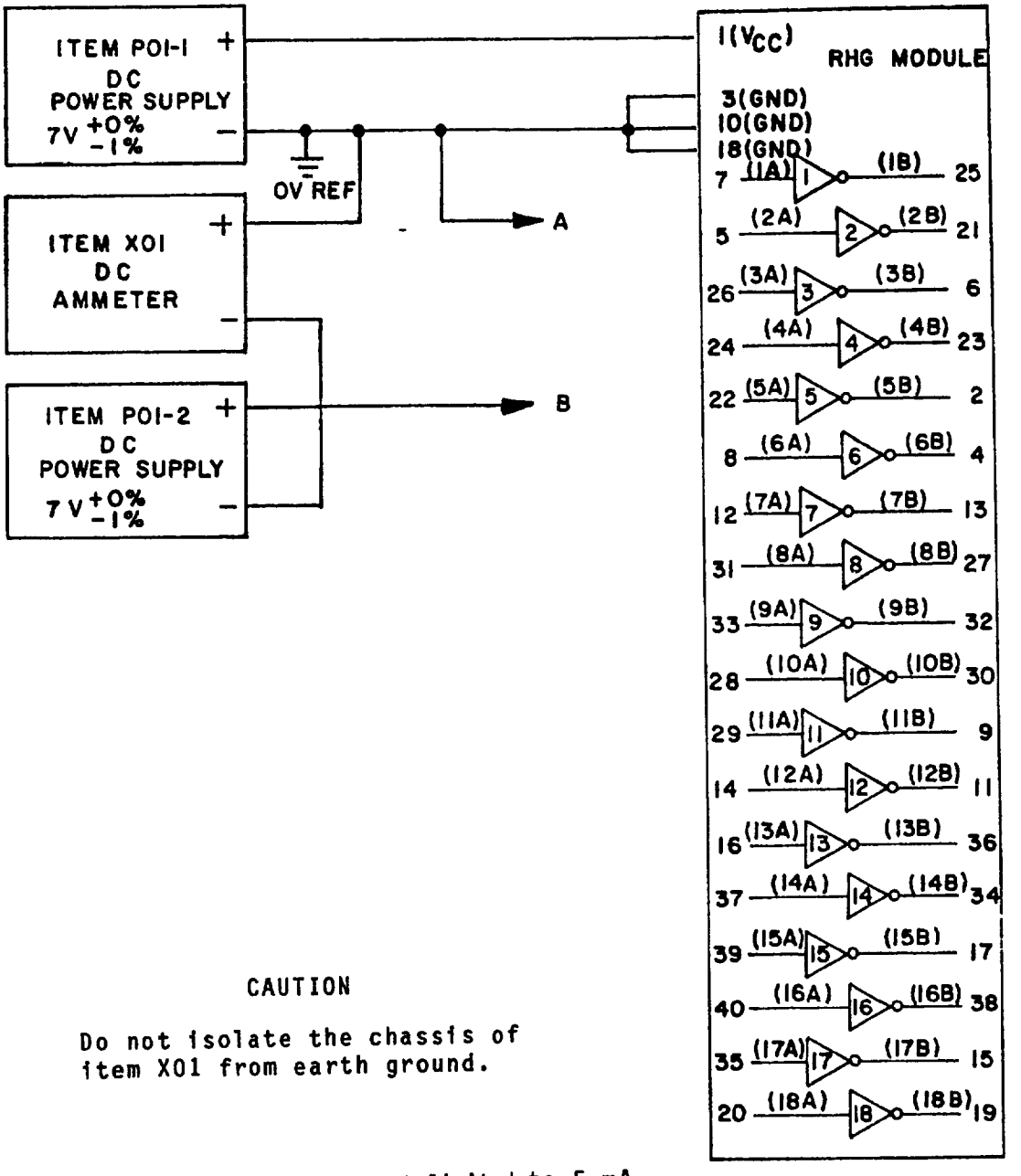
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1A	Input breakdown current	5,8,12,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	7	0	100	µA	7
C2	2A	"	7,8,12,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	5	"	"	"	"
C3	3A	"	5,7,8,12,14,16,20,22,24,28, 29,31,33,35,37,39,40	26	"	"	"	"
C4	4A	"	5,7,8,12,14,16,20,22,26,28, 29,31,33,35,37,39,40	24	"	"	"	"
C5	5A	"	5,7,8,12,14,16,20,24,26,28, 29,31,33,35,37,39,40	22	"	"	"	"
C6	6A	"	5,7,12,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	8	"	"	"	"
C7	7A	"	5,7,8,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	12	"	"	"	"
C8	8A	"	5,7,8,12,14,16,20,22,24,26, 28,29,33,35,37,39,40	31	"	"	"	"
C9	9A	"	5,7,8,12,14,16,20,22,24,26, 28,29,31,35,37,39,40	33	"	"	"	"
C10	10A	"	5,7,8,12,14,16,20,22,24,26, 29,31,33,35,37,39,40	28	"	"	"	"
C11	11A	"	5,7,8,12,14,16,20,22,24,26, 28,31,33,35,37,39,40	29	"	"	"	"
C12	12A	"	5,7,8,12,16,20,22,24,26, 28,29,31,33,35,37,39,40	14	"	"	"	"
C13	13A	"	5,7,8,12,14,20,22,24,26, 28,29,31,33,35,37,39,40	16	"	"	"	"
C14	14A	"	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,39,40	37	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C15	15A Input breakdown current	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,40	39	0	100	µA	7
C16	16A "	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39	40	"	"	"	"
C17	17A "	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,37,39,40	35	"	"	"	"
C18	18A "	5,7,8,12,14,16,22,24,26, 28,29,31,33,35,37,39,40	20	"	"	"	"

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CAUTION

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Item PO1-2 shall be current limited to 5 mA.

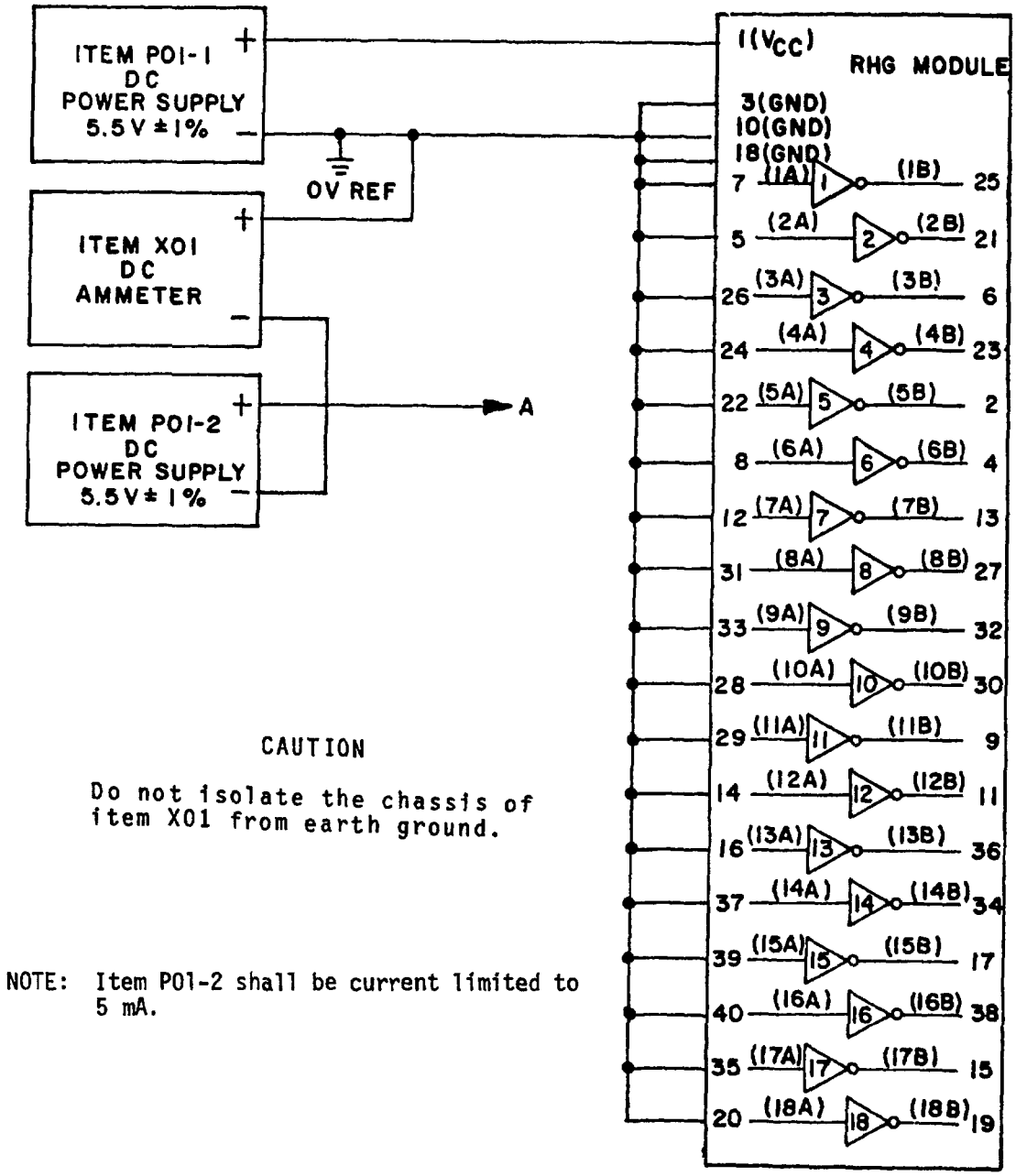
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1B Output leakage current	25	0	1	mA	8
D2	2B "	21	"	"	"	"
D3	3B "	6	"	"	"	"
D4	4B "	23	"	"	"	"
D5	5B "	2	"	"	"	"
D6	6B "	4	"	"	"	"
D7	7B "	13	"	"	"	"
D8	8B "	27	"	"	"	"
D9	9B "	32	"	"	"	"
D10	10B "	30	"	"	"	"
D11	11B "	9	"	"	"	"
D12	12B "	11	"	"	"	"
D13	13B "	36	"	"	"	"
D14	14B "	34	"	"	"	"
D15	15B "	17	"	"	"	"
D16	16B "	38	"	"	"	"
D17	17B "	15	"	"	"	"
D18	18B "	19	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5 mA.

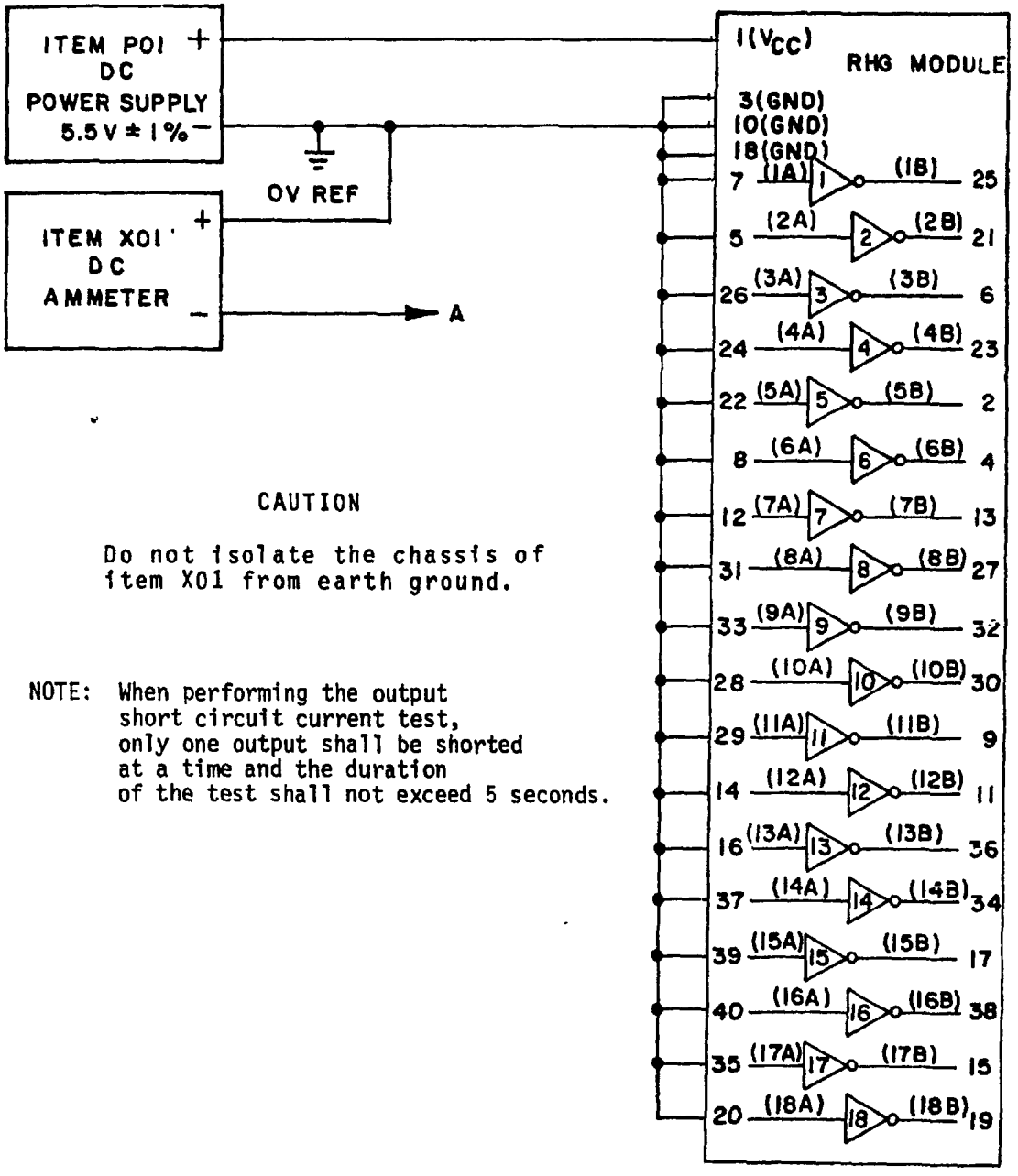
FIGURE 8. Output leakage current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1B Output short circuit current	25	-15	-100	mA	9
E2	2B "	21	"	"	"	"
E3	3B "	6	"	"	"	"
E4	4B "	23	"	"	"	"
E5	5B "	2	"	"	"	"
E6	6B "	4	"	"	"	"
E7	7B "	13	"	"	"	"
E8	8B "	27	"	"	"	"
E9	9B "	32	"	"	"	"
E10	10B "	30	"	"	"	"
E11	11B "	9	"	"	"	"
E12	12B "	11	"	"	"	"
E13	13B "	36	"	"	"	"
E14	14B "	34	"	"	"	"
E15	15B "	17	"	"	"	"
E16	16B "	38	"	"	"	"
E17	17B "	15	"	"	"	"
E18	18B "	19	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

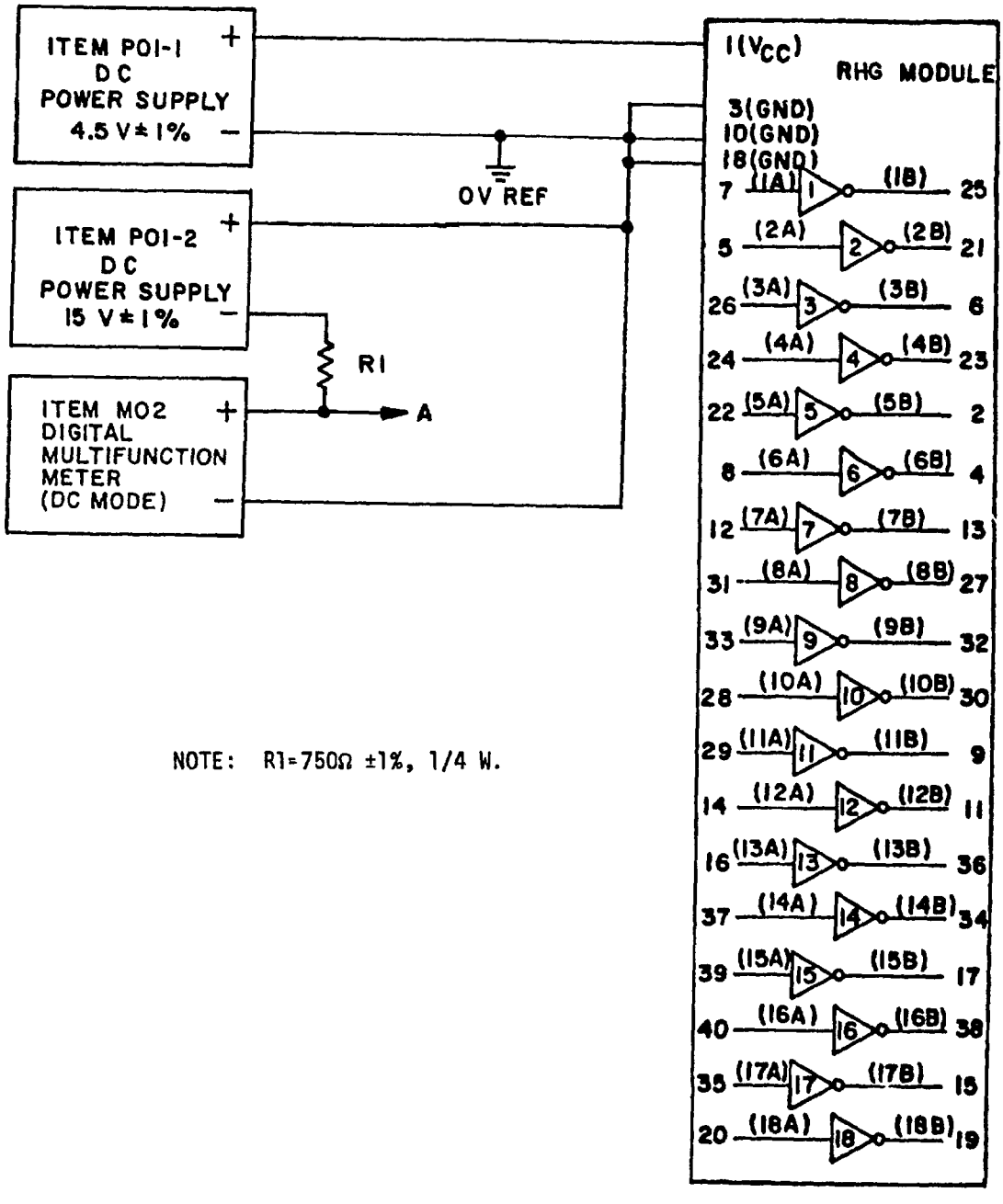
FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	1A Input clamp diode voltage	7	-0.4	-1.5	V	10
F2	2A "	5	"	"	"	"
F3	3A "	26	"	"	"	"
F4	4A "	24	"	"	"	"
F5	5A "	22	"	"	"	"
F6	6A "	8	"	"	"	"
F7	7A "	12	"	"	"	"
F8	8A "	31	"	"	"	"
F9	9A "	33	"	"	"	"
F10	10A "	28	"	"	"	"
F11	11A "	29	"	"	"	"
F12	12A "	14	"	"	"	"
F13	13A "	16	"	"	"	"
F14	14A "	37	"	"	"	"
F15	15A "	39	"	"	"	"
F16	16A "	40	"	"	"	"
F17	17A "	35	"	"	"	"
F18	18A "	20	"	"	"	"

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NOTE: R1=750Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Output voltage (low-level) test.

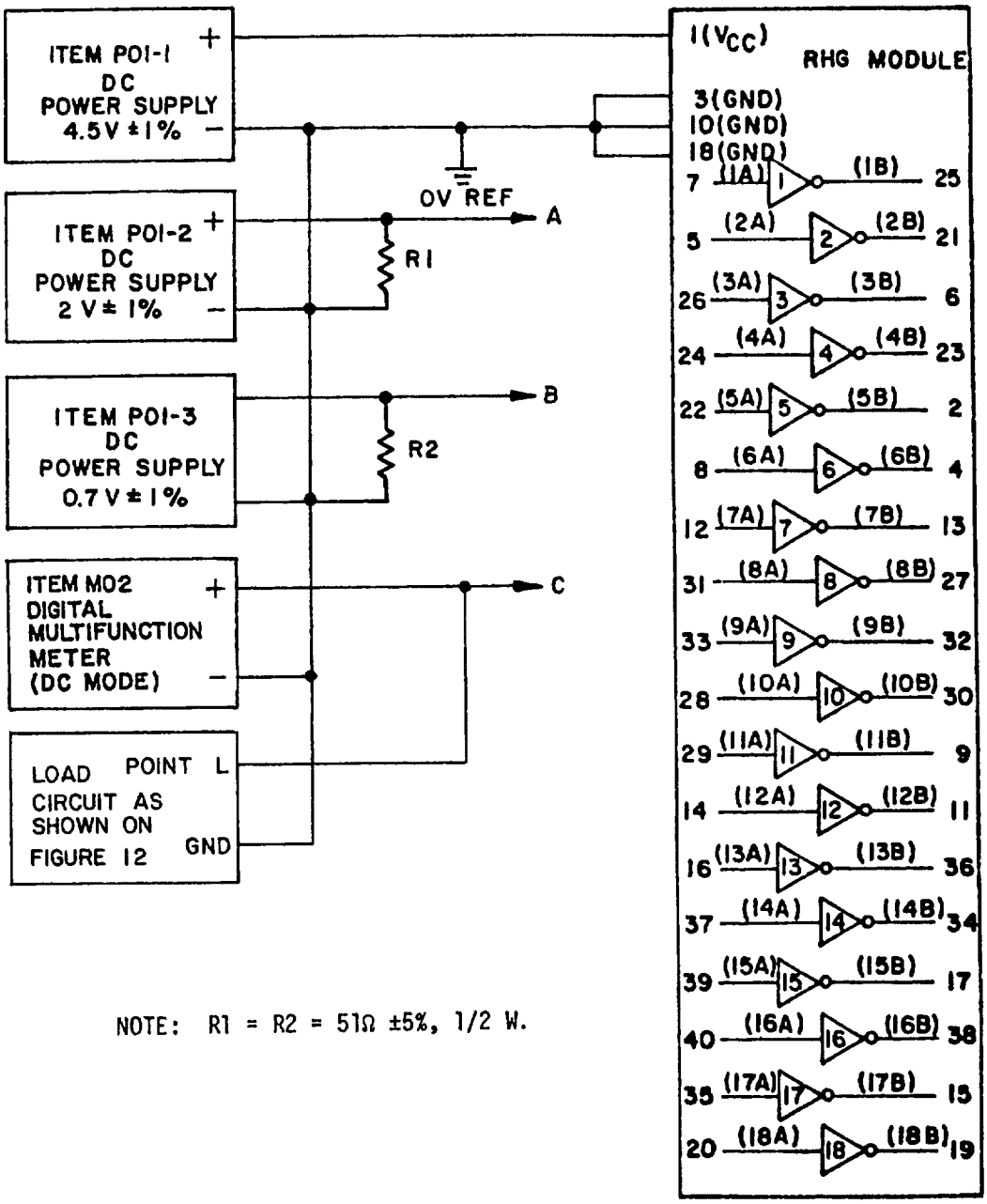
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1B Output voltage, low level	7	Open	25	0	400	mV	11
G2	2B "	5	"	21	"	"	"	"
G3	3B "	26	"	6	"	"	"	"
G4	4B "	24	"	23	"	"	"	"
G5	5B "	22	"	2	"	"	"	"
G6	6B "	8	"	4	"	"	"	"
G7	7B "	12	"	13	"	"	"	"
G8	8B "	31	"	27	"	"	"	"
G9	9B "	33	"	32	"	"	"	"
G10	10B "	28	"	30	"	"	"	"
G11	11B "	29	"	9	"	"	"	"
G12	12B "	14	"	11	"	"	"	"
G13	13B "	16	"	36	"	"	"	"
G14	14B "	37	"	34	"	"	"	"
G15	15B "	39	"	17	"	"	"	"
G16	16B "	40	"	38	"	"	"	"
G17	17B "	35	"	15	"	"	"	"
G18	18B "	20	"	19	"	"	"	"

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TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Output voltage, high level	Open	7	25	2.5	4.5	V	11
H2	2B "	"	5	21	"	"	"	"
H3	3B "	"	26	6	"	"	"	"
H4	4B "	"	24	23	"	"	"	"
H5	5B "	"	22	2	"	"	"	"
H6	6B "	"	8	4	"	"	"	"
H7	7B "	"	12	13	"	"	"	"
H8	8B "	"	31	27	"	"	"	"
H9	9B "	"	33	32	"	"	"	"
H10	10B "	"	28	30	"	"	"	"
H11	11B "	"	29	9	"	"	"	"
H12	12B "	"	14	11	"	"	"	"
H13	13B "	"	16	36	"	"	"	"
H14	14B "	"	37	34	"	"	"	"
H15	15B "	"	39	17	"	"	"	"
H16	16B "	"	40	38	"	"	"	"
H17	17B "	"	35	15	"	"	"	"
H18	18B "	"	20	19	"	"	"	"

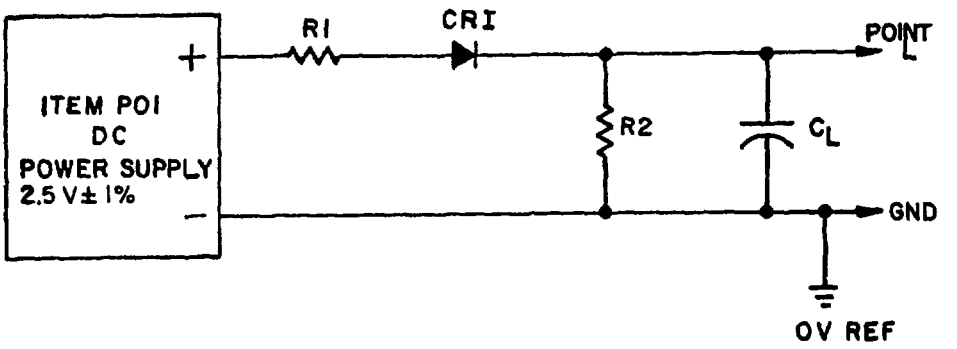
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

*FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 4 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 6.19 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for test conditions G1-G18 and H1-H18.
6. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 12. RHG logic load circuit.

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TABLE XI. Input current (high-level) test.

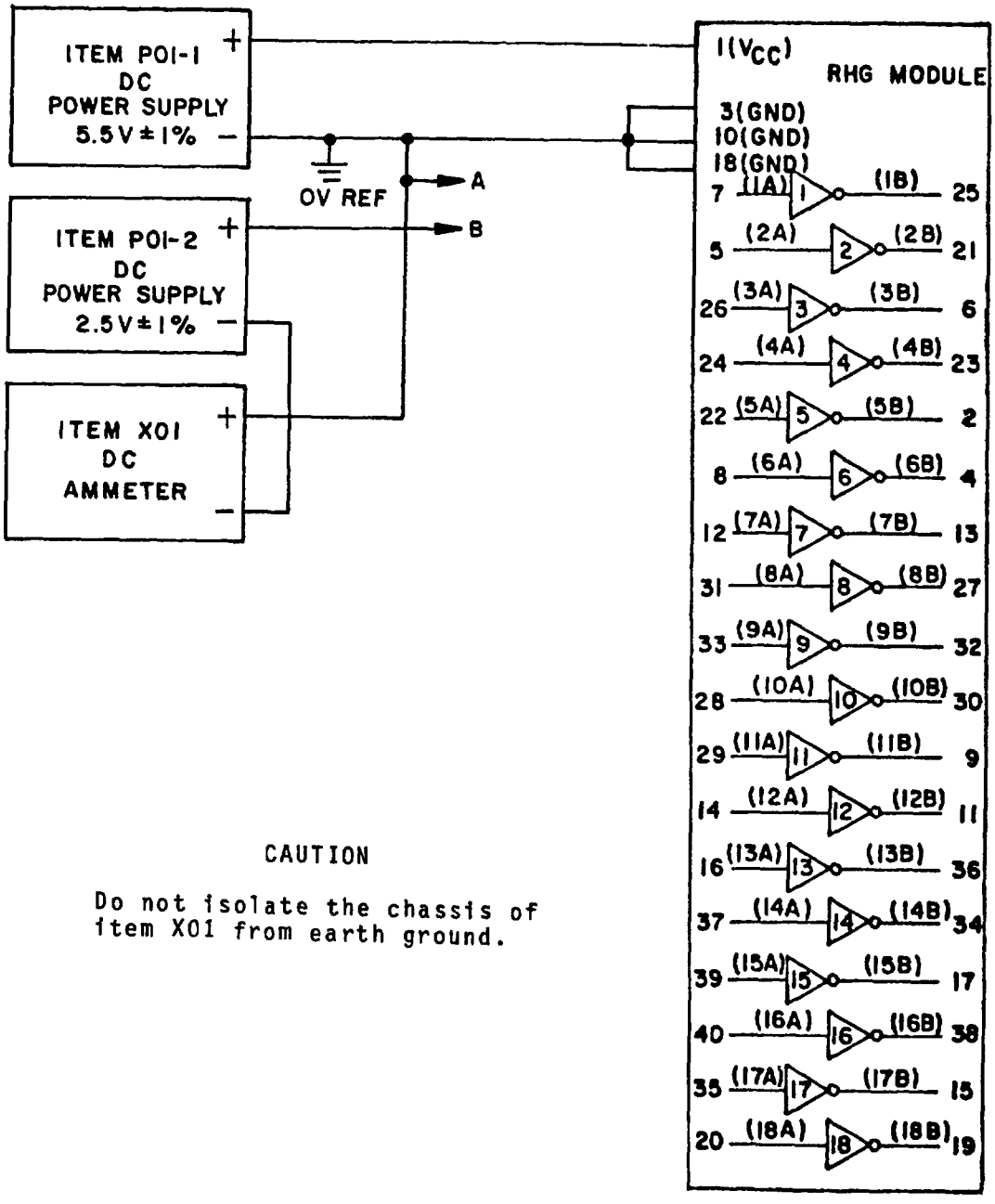
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A Input current, high level	5,8,12,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	7	0	20	µA	13
J2	2A "	7,8,12,14,16,20,22,24,26,28, 29,31,33,35,37,39,40	5	"	"	"	"
J3	3A "	5,7,8,12,14,16,20,22,24,28, 29,31,33,35,37,39,40	26	"	"	"	"
J4	4A "	5,7,8,12,14,16,20,22,26,28, 29,31,33,35,37,39,40	24	"	"	"	"
J5	5A "	5,7,8,12,14,16,20,24,26, 28,29,31,33,35,37,39,40	22	"	"	"	"
J6	6A "	5,7,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	8	"	"	"	"
J7	7A "	5,7,8,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	12	"	"	"	"
J8	8A "	5,7,8,12,14,16,20,22,24,26, 28,29,33,35,37,39,40	31	"	"	"	"
J9	9A "	5,7,8,12,14,16,20,22,24,26, 28,29,31,35,37,39,40	33	"	"	"	"
J10	10A "	5,7,8,12,14,16,20,22,24,26, 29,31,33,35,37,39,40	28	"	"	"	"
J11	11A "	5,7,8,12,14,16,20,22,24,26, 28,31,33,35,37,39,40	29	"	"	"	"
J12	12A "	5,7,8,12,16,20,22,24,26, 28,29,31,33,35,37,39,40	14	"	"	"	"
J13	13A "	5,7,8,12,14,20,22,24,26, 28,29,31,33,35,37,39,40	16	"	"	"	"
J14	14A "	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,39,40	37	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J15	15A Input current, high level	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,40	39	0	20	µA	13
J16	16A "	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39	40	"	"	"	"
J17	17A "	5,7,8,12,14,16,20,22,24,26, 28,29,31,33,37,39,40	35	"	"	"	"
J18	18A "	5,7,8,12,14,16,22,24,26, 28,29,31,33,35,37,39,40	20	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

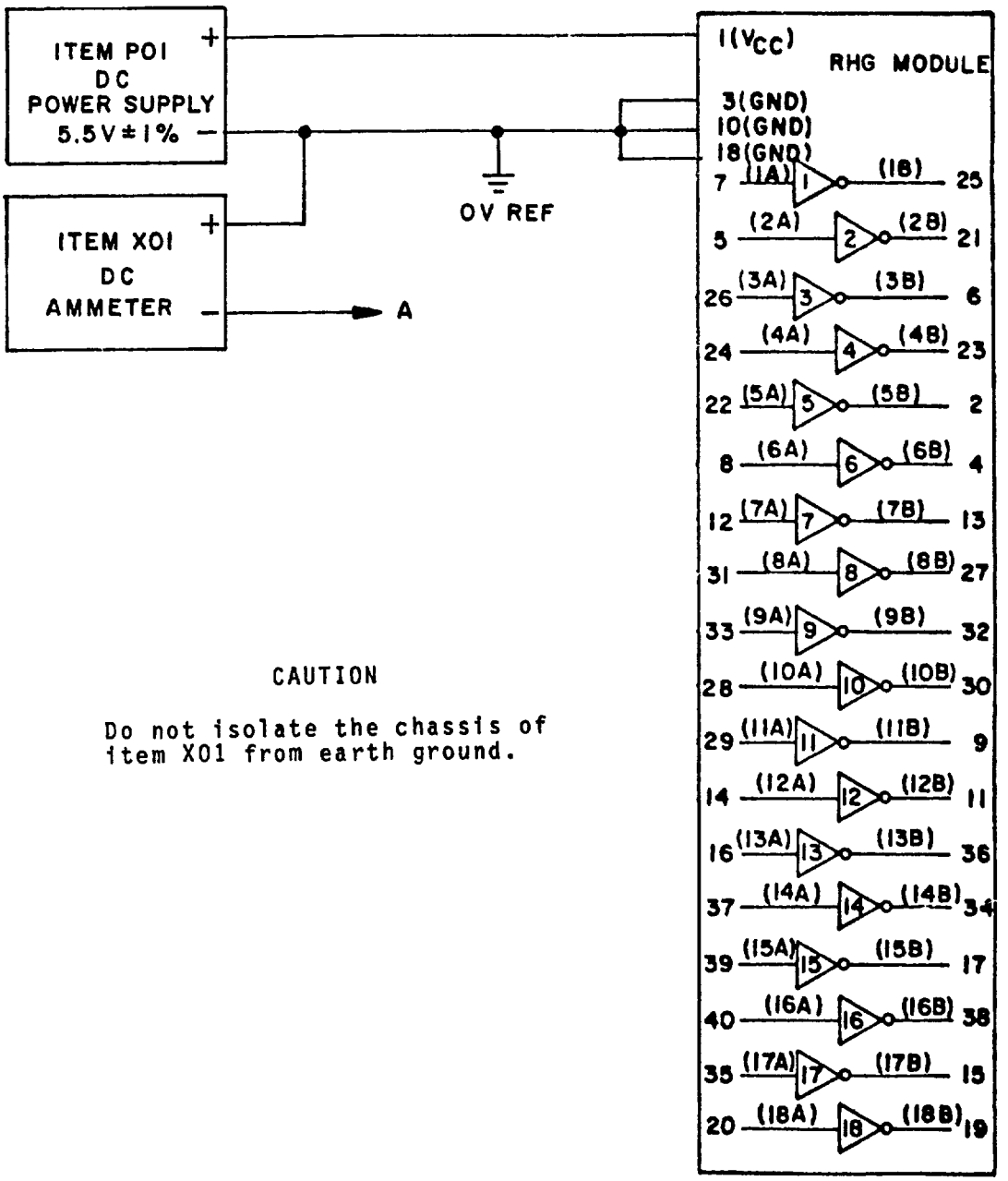
FIGURE 13. Input current (high-level) test circuit for table XI.

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*TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1A Input current, low level	7	-10	-430	µA	14
K2	2A "	5	"	"	"	"
K3	3A "	26	"	"	"	"
K4	4A "	24	"	"	"	"
K5	5A "	22	"	"	"	"
K6	6A "	8	"	"	"	"
K7	7A "	12	"	"	"	"
K8	8A "	31	"	"	"	"
K9	9A "	33	"	"	"	"
K10	10A "	28	"	"	"	"
K11	11A "	29	"	"	"	"
K12	12A "	14	"	"	"	"
K13	13A "	16	"	"	"	"
K14	14A "	37	"	"	"	"
K15	15A "	39	"	"	"	"
K16	16A "	40	"	"	"	"
K17	17A "	35	"	"	"	"
K18	18A "	20	"	"	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

FIGURE 14. Input current (low-level) test circuit for table XII.

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TABLE XIII. Ground pin continuity test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Ground pin continuity	3	0	1	Ω	15
L2	"	18	0	1	"	"
L3	"	3	0.00	0.10	"	"
L4	"	18	0.00	0.10	"	"

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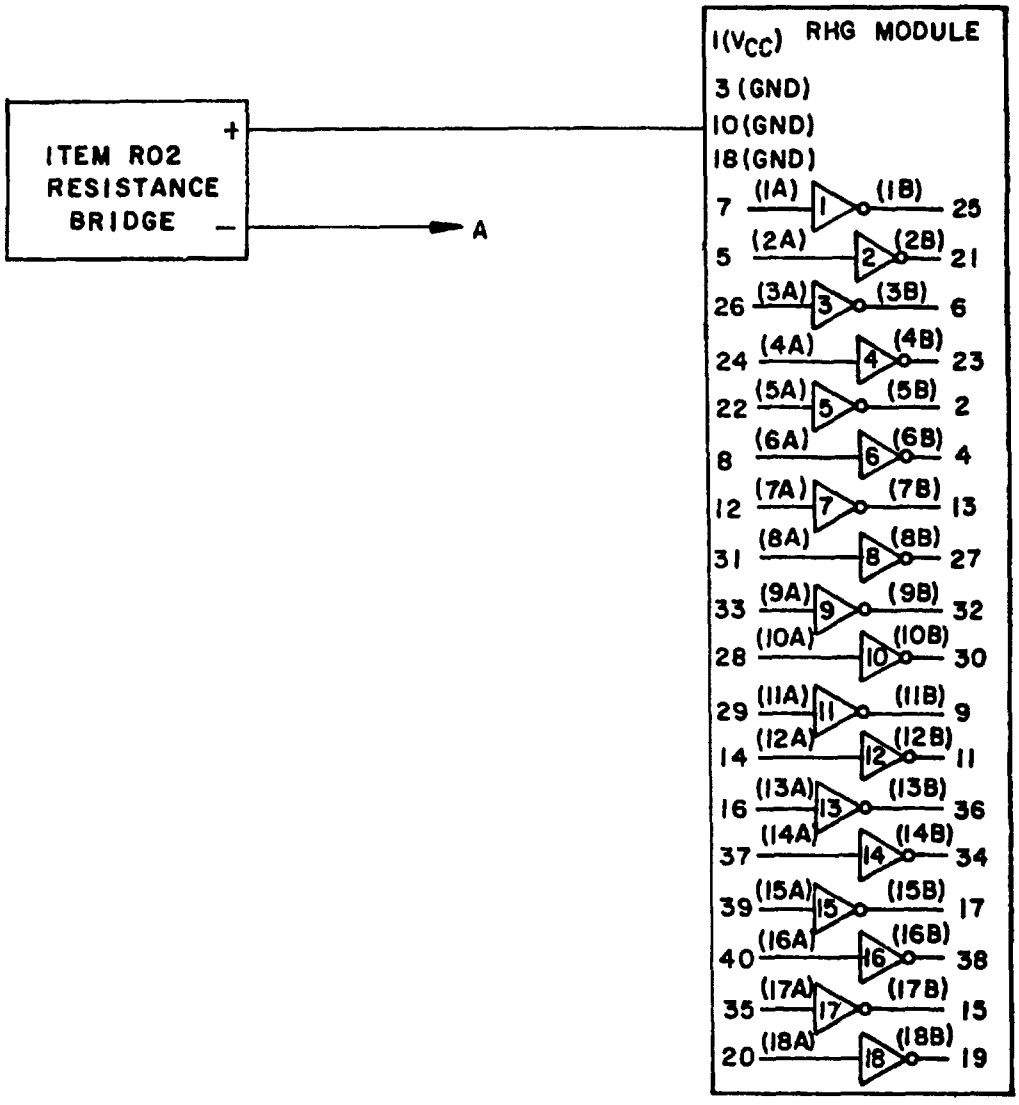


FIGURE 15. Ground pin continuity test circuit for table XIII.

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	C	Open	0.1	19.8	mA	16
M2	Power supply current, high level	Open	C	0.1	7.2	mA	16

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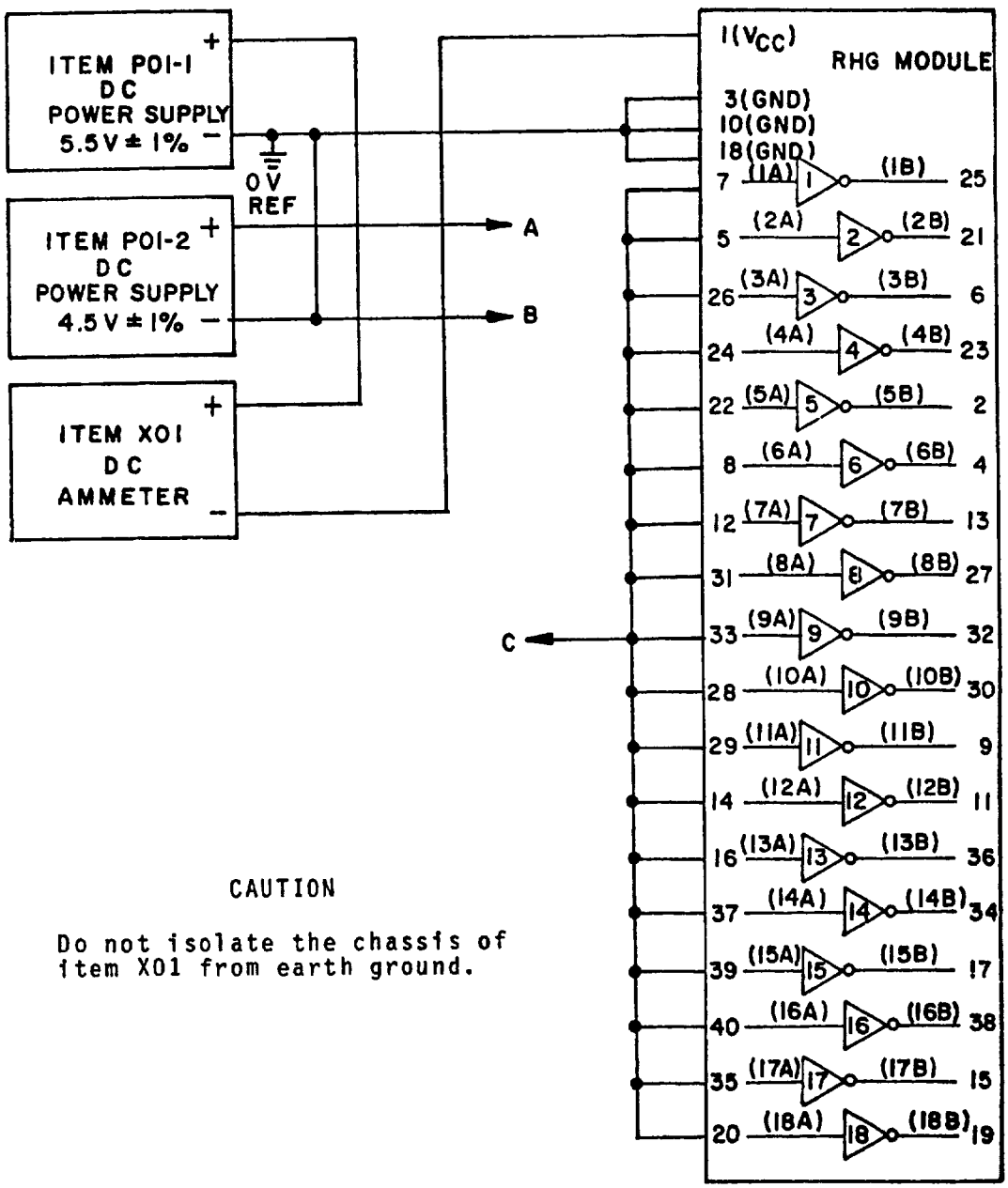


FIGURE 16. Power supply current (low-level and high-level) test circuit for table XIV.

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N1	1B	Transition time (t_{TLH}), C_L = 50 pF	7	25	5	50	ns	17,18
N2	2B	"	5	21	"	"	"	"
N3	3B	"	26	6	"	"	"	"
N4	4B	"	24	23	"	"	"	"
N5	5B	"	22	2	"	"	"	"
N6	6B	"	8	4	"	"	"	"
N7	7B	"	12	13	"	"	"	"
N8	8B	"	31	27	"	"	"	"
N9	9B	"	33	32	"	"	"	"
N10	10B	"	28	30	"	"	"	"
N11	11B	"	29	9	"	"	"	"
N12	12B	"	14	11	"	"	"	"
N13	13B	"	16	36	"	"	"	"
N14	14B	"	37	34	"	"	"	"
N15	15B	"	39	17	"	"	"	"
N16	16B	"	40	38	"	"	"	"
N17	17B	"	35	15	"	"	"	"
N18	18B	"	20	19	"	"	"	"
N19	1B	Transition time (t_{TLH}), C_L = 150 pF	7	25	10	80	"	"
N20	2B	"	5	21	"	"	"	"
N21	3B	"	26	6	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N22	4B Transition time (t_{TLH}), C_L = 150 pF	24	23	10	80	ns	17,18
N23	5B "	22	2	"	"	"	"
N24	6B "	8	4	"	"	"	"
N25	7B "	12	13	"	"	"	"
N26	8B "	31	27	"	"	"	"
N27	9B "	33	32	"	"	"	"
N28	10B "	28	30	"	"	"	"
N29	11B "	29	9	"	"	"	"
N30	12B "	14	11	"	"	"	"
N31	13B "	16	36	"	"	"	"
N32	14B "	37	34	"	"	"	"
N33	15B "	39	17	"	"	"	"
N34	16B "	40	38	"	"	"	"
N35	17B "	35	15	"	"	"	"
N36	18B "	20	19	"	"	"	"
P1	1B Transition time (t_{THL}), C_L = 50 pF	7	25	3	18	"	"
P2	2B "	5	21	"	"	"	"
P3	3B "	26	6	"	"	"	"
P4	4B "	24	23	"	"	"	"
P5	5B "	22	2	"	"	"	"
P6	6B "	8	4	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P7	7B Transition time (t_{THL}), C_L = 50 pF	12	13	3	18	ns	17,18
P8	8B "	31	27	"	"	"	"
P9	9B "	33	32	"	"	"	"
P10	10B "	28	30	"	"	"	"
P11	11B "	29	9	"	"	"	"
P12	12B "	14	11	"	"	"	"
P13	13B "	16	36	"	"	"	"
P14	14B "	37	34	"	"	"	"
P15	15B "	39	17	"	"	"	"
P16	16B "	40	38	"	"	"	"
P17	17B "	35	15	"	"	"	"
P18	18B "	20	19	"	"	"	"
P19	1B Transition time (t_{THL}), C_L = 150 pF	7	25	6	40	"	"
P20	2B "	5	21	"	"	"	"
P21	3B "	26	6	"	"	"	"
P22	4B "	24	23	"	"	"	"
P23	5B "	22	2	"	"	"	"
P24	6B "	8	4	"	"	"	"
P25	7B "	12	13	"	"	"	"
P26	8B "	31	27	"	"	"	"
P27	9B "	33	32	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P28	10B Transition time (t_{THL}), C_L = 150 pF	28	30	6	40	ns	17,18
P29	11B "	29	9	"	"	"	"
P30	12B "	14	11	"	"	"	"
P31	13B "	16	36	"	"	"	"
P32	14B "	37	34	"	"	"	"
P33	15B "	39	17	"	"	"	"
P34	16B "	40	38	"	"	"	"
P35	17B "	35	15	"	"	"	"
P36	18B "	20	19	"	"	"	"

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	1B Propagation delay time (t_{PLH}), C_L = 50 pF	7	25	3	19	ns	17,18
Q2	2B "	5	21	"	"	"	"
Q3	3B "	26	6	"	"	"	"
Q4	4B "	24	23	"	"	"	"
Q5	5B "	22	2	"	"	"	"
Q6	6B "	8	4	"	"	"	"
Q7	7B "	12	13	"	"	"	"
Q8	8B "	31	27	"	"	"	"
Q9	9B "	33	32	"	"	"	"
Q10	10B "	28	30	"	"	"	"
Q11	11B "	29	9	"	"	"	"
Q12	12B "	14	11	"	"	"	"
Q13	13B "	16	36	"	"	"	"
Q14	14B "	37	34	"	"	"	"
Q15	15B "	39	17	"	"	"	"
Q16	16B "	40	38	"	"	"	"
Q17	17B "	35	15	"	"	"	"
Q18	18B "	20	19	"	"	"	"
Q19	1B Propagation delay time (t_{PLH}), C_L = 150 pF	7	25	4	29	"	"
Q20	2B "	5	21	"	"	"	"
Q21	3B "	26	6	"	"	"	"

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q22	4B Propagation delay time (t_{PLH}), C_L = 150 pF	24	23	4	29	ns	17,18
Q23	5B "	22	2	"	"	"	"
Q24	6B "	8	4	"	"	"	"
Q25	7B "	12	13	"	"	"	"
Q26	8B "	31	27	"	"	"	"
Q27	9B "	33	32	"	"	"	"
Q28	10B "	28	30	"	"	"	"
Q29	11B "	29	9	"	"	"	"
Q30	12B "	14	11	"	"	"	"
Q31	13B "	16	36	"	"	"	"
Q32	14B "	37	34	"	"	"	"
Q33	15B "	39	17	"	"	"	"
Q34	16B "	40	38	"	"	"	"
Q35	17B "	35	15	"	"	"	"
Q36	18B "	20	19	"	"	"	"
R1	1B Propagation delay time (t_{PHL}), C_L = 50 pF	7	25	3	19	"	"
R2	2B "	5	21	"	"	"	"
R3	3B "	26	6	"	"	"	"
R4	4B "	24	23	"	"	"	"
R5	5B "	22	2	"	"	"	"
R6	6B "	8	4	"	"	"	"

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

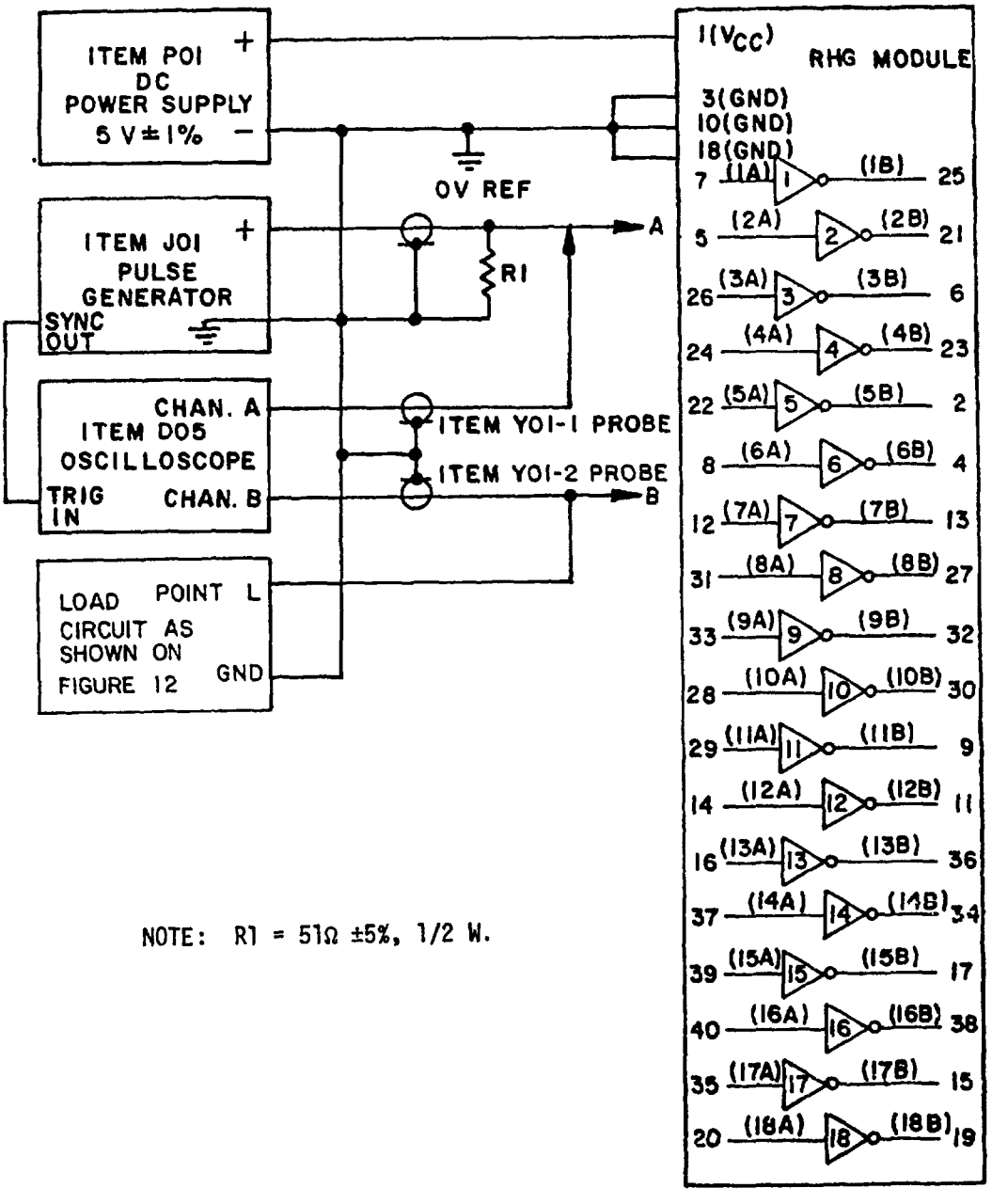
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R7	7B Propagation delay time (t_{PHL}), C_L = 50 pF	12	13	3	19	ns	17,18
R8	8B "	31	27	"	"	"	"
R9	9B "	33	32	"	"	"	"
R10	10B "	28	30	"	"	"	"
R11	11B "	29	9	"	"	"	"
R12	12B "	14	11	"	"	"	"
R13	13B "	16	36	"	"	"	"
R14	14B "	37	34	"	"	"	"
R15	15B "	39	17	"	"	"	"
R16	16B "	40	38	"	"	"	"
R17	17B "	35	15	"	"	"	"
R18	18B "	20	19	"	"	"	"
R19	1B Propagation delay time (t_{PHL}), C_L = 150 pF	7	25	4	29	"	"
R20	2B "	5	21	"	"	"	"
R21	3B "	26	6	"	"	"	"
R22	4B "	24	23	"	"	"	"
R23	5B "	22	2	"	"	"	"
R24	6B "	8	4	"	"	"	"
R25	7B "	12	13	"	"	"	"
R26	8B "	31	27	"	"	"	"

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*TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
R27	9B	Propagation delay time (t_{PHL}), C_L = 150 pF	33	32	4	29	ns	17,18
R28	10B	"	28	30	"	"	"	"
R29	11B	"	29	9	"	"	"	"
R30	12B	"	14	11	"	"	"	"
R31	13B	"	16	36	"	"	"	"
R32	14B	"	37	34	"	"	"	"
R33	15B	"	39	17	"	"	"	"
R34	16B	"	40	38	"	"	"	"
R35	17B	"	35	15	"	"	"	"
R36	18B	"	20	19	"	"	"	"

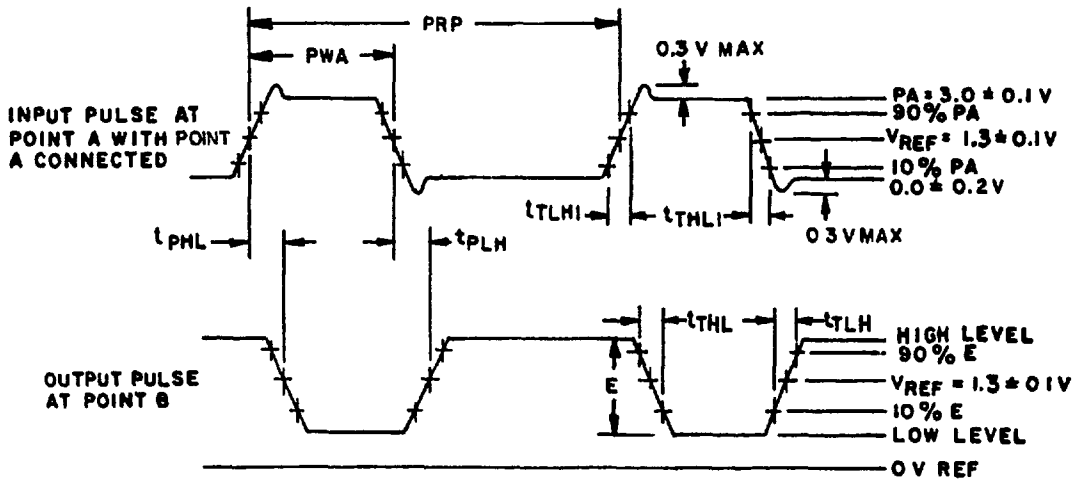
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NOTE: R1 = 51Ω ±5%, 1/2 W.

*FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pin; between items YO1-1 and YO1-2 and input and output pins; and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

*FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1B Output crosstalk, low level	7	Open	5,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	25	0	650	mV	19,20
S2	2B "	5	"	7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	21	"	"	"	"
S3	3B "	26	"	5,7,8,12,14,16,20,22,24, 28,29,31,33,35,37,39,40	6	"	"	"	"
S4	4B "	24	"	5,7,8,12,14,16,20,22,26, 28,29,31,33,35,37,39,40	23	"	"	"	"
S5	5B "	22	"	5,7,8,12,14,16,20,24,26, 28,29,31,33,35,37,39,40	2	"	"	"	"
S6	6B "	8	"	5,7,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	4	"	"	"	"
S7	7B "	12	"	5,7,8,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	13	"	"	"	"
S8	8B "	31	"	5,7,8,12,14,16,20,22,24, 26,28,29,33,35,37,39,40	27	"	"	"	"
S9	9B "	33	"	5,7,8,12,14,16,20,22,24, 26,28,29,31,35,37,39,40	32	"	"	"	"
S10	10B "	28	"	5,7,8,12,14,16,20,22,24, 26,29,31,33,35,37,39,40	30	"	"	"	"
S11	11B "	29	"	5,7,8,12,14,16,20,22,24, 26,28,31,33,35,37,39,40	9	"	"	"	"
S12	12B "	14	"	5,7,8,12,16,20,22,24,26, 28,29,31,33,35,37,39,40	11	"	"	"	"
S13	13B "	16	"	5,7,8,12,14,20,22,24,26, 28,29,31,33,35,37,39,40	36	"	"	"	"
S14	14B "	37	"	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,39,40	34	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S15	15B Output crosstalk, low level	39	Open	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,40	17	0	650	mV	19,20
S16	16B "	40	"	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,39	38	"	"	"	"
S17	17B "	35	"	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,37,39,40	15	"	"	"	"
S18	18B "	20	"	5,7,8,12,14,16,22,24,26, 28,29,31,33,35,37,39,40	19	"	"	"	"
S19	1B "	5,7,26,8, 22,24	"	12,14,16,20,28,29,31,33, 35,37,39,40	25	"	400	"	"
S20	2B "	"	"	"	21	"	"	"	"
S21	3B "	"	"	"	6	"	"	"	"
S22	4B "	"	"	"	23	"	"	"	"
S23	5B "	"	"	"	2	"	"	"	"
S24	6B "	"	"	"	4	"	"	"	"
S25	7B "	12,31,33, 14,28,29	"	5,7,8,16,20,22,24,26,35, 37,39,40	13	"	"	"	"
S26	8B "	"	"	"	27	"	"	"	"
S27	9B "	"	"	"	32	"	"	"	"
S28	10B "	"	"	"	30	"	"	"	"
S29	11B "	"	"	"	9	"	"	"	"
S30	12B "	"	"	"	11	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

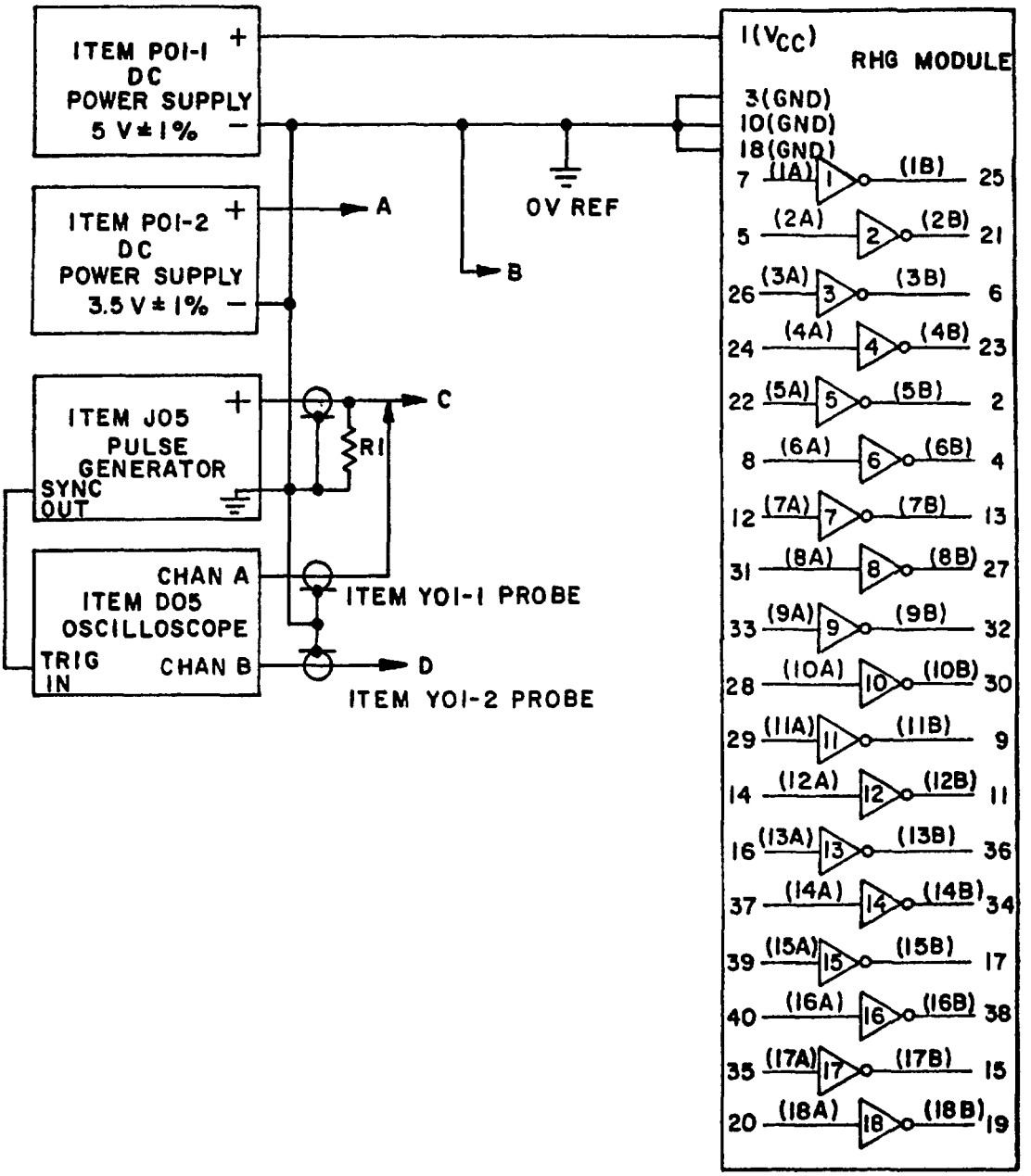
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S31	13B Output crosstalk, low level	16,37,39, 20,35,40	Open	5,7,8,12,14,22,24,26,28, 29,31,33	36	0	400	mV	19,20
S32	14B "	"	"	"	34	"	"	"	"
S33	15B "	"	"	"	17	"	"	"	"
S34	16B "	"	"	"	38	"	"	"	"
S35	17B "	"	"	"	15	"	"	"	"
S36	18B "	"	"	"	19	"	"	"	"
T1	1B Output crosstalk, high level	Open	7	5,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	25	"	"	"	"
T2	2B "	"	5	7,8,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	21	"	"	"	"
T3	3B "	"	26	5,7,8,12,14,16,20,22,24, 28,29,31,33,35,37,39,40	6	"	"	"	"
T4	4B "	"	24	5,7,8,12,14,16,20,22,26, 28,29,31,33,35,37,39,40	23	"	"	"	"
T5	5B "	"	22	5,7,8,12,14,16,20,24,26, 28,29,31,33,35,37,39,40	2	"	"	"	"
T6	6B "	"	8	5,7,12,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	4	"	"	"	"
T7	7B "	"	12	5,7,8,14,16,20,22,24,26, 28,29,31,33,35,37,39,40	13	"	"	"	"
T8	8B "	"	31	5,7,8,12,14,16,20,22,24, 26,28,29,33,35,37,39,40	27	"	"	"	"
T9	9B "	"	33	5,7,8,12,14,16,20,22,24, 26,28,29,31,35,37,39,40	32	"	"	"	"
T10	10B "	"	28	5,7,8,12,14,16,20,22,24, 26,29,31,33,35,37,39,40	30	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T11	11B Output crosstalk, high level	Open	29	5,7,8,12,14,16,20,22,24, 26,28,31,33,35,37,39,40	9	0	400	mV	19,20
T12	12B "	"	14	5,7,8,12,16,20,22,24,26, 28,29,31,33,35,37,39,40	11	"	"	"	"
T13	13B "	"	16	5,7,8,12,14,20,22,24,26, 28,29,31,33,35,37,39,40	36	"	"	"	"
T14	14B "	"	37	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,39,40	34	"	"	"	"
T15	15B "	"	39	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,40	17	"	"	"	"
T16	16B "	"	40	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,35,37,39	38	"	"	"	"
T17	17B "	"	35	5,7,8,12,14,16,20,22,24, 26,28,29,31,33,37,39,40	15	"	"	"	"
T18	18B "	"	20	5,7,8,12,14,16,22,24,26, 28,29,31,33,35,37,39,40	19	"	"	"	"

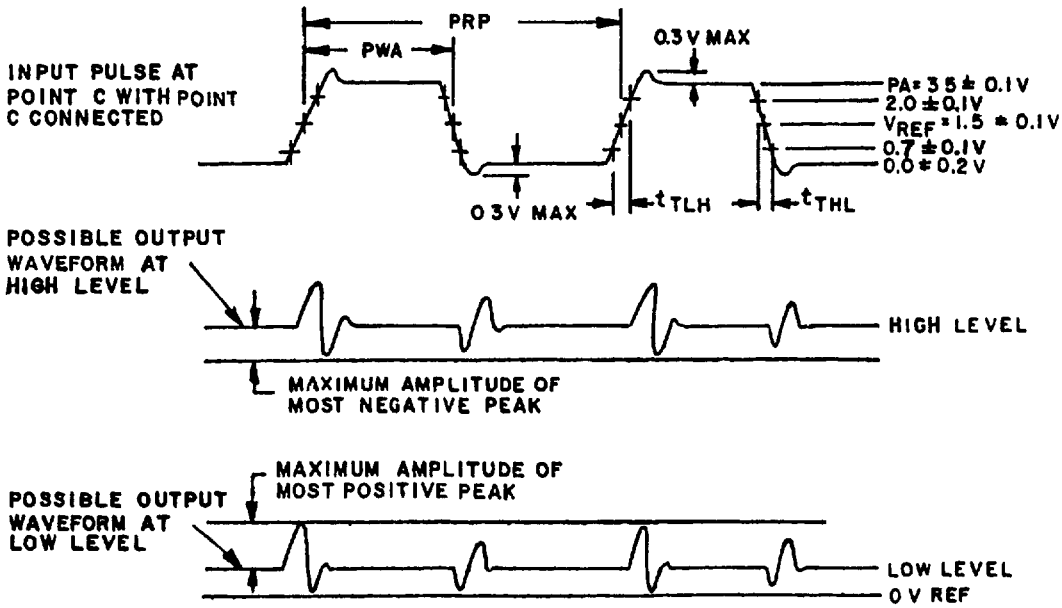
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 12 with the exception that C_L = 50 ±5 pF.

FIGURE 19. Output crosstalk (low-level and high-level) test circuit for table XVII.

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NOTES:
1. Input pulse parameters: For each test condition with the module connected to item J05, adjust item J05 for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parameters at each input pin before making each measurement.

   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (measured at the 0.7 V and 2.0 V points of the input pulse).
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of item Y01-1 and Y01-2 and test fixture and module ground pins shall be less than 1 inch (25.4 mm).

*FIGURE 20. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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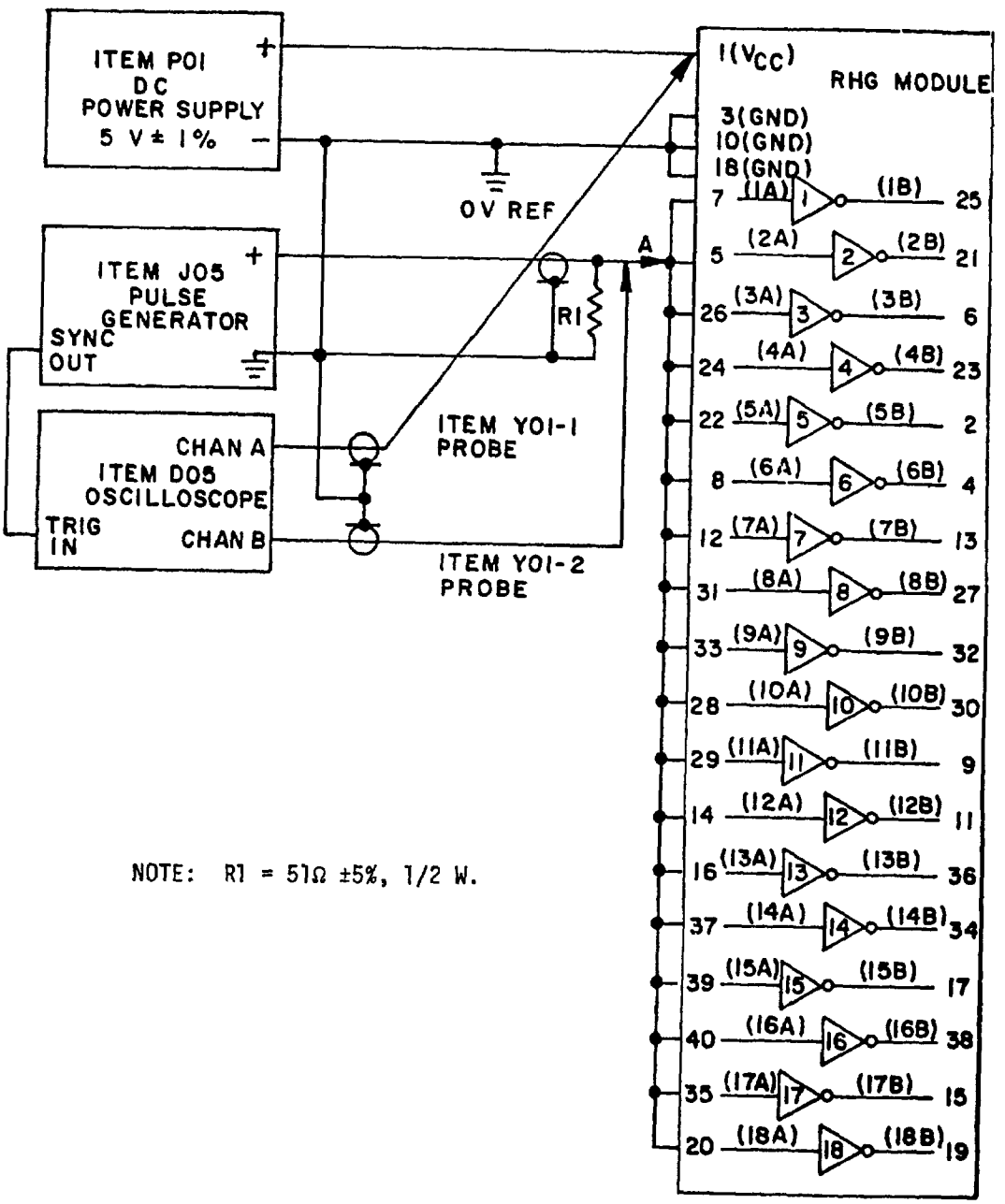
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	500	mV	21,22

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	23

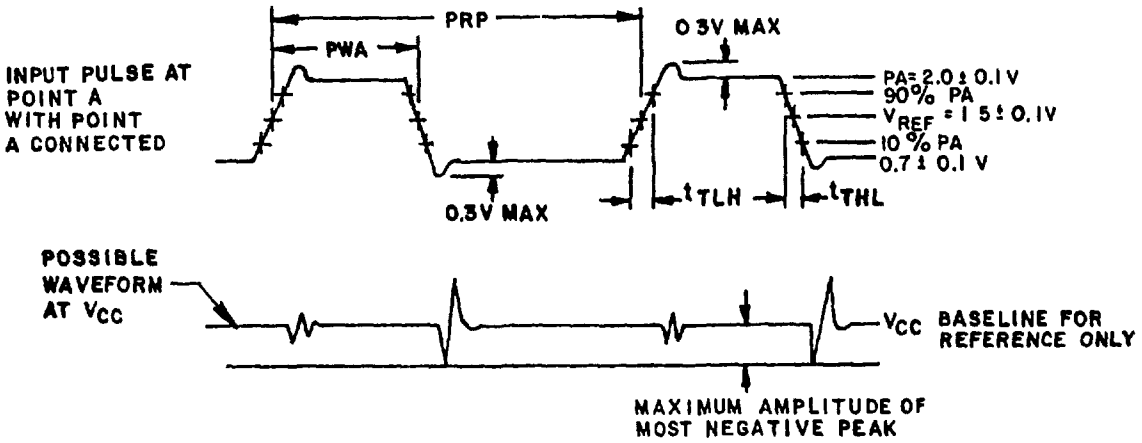
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 21. Power supply transient amplitude test circuit for table XVIII.

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NOTE: Input pulse parameters: With the module connected to item J05, adjust item J05 for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parameters at each input pin before making the measurement.
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns verified at each input pin.
PWA = 500 ±50 ns.
PRP = 1,000 ±200 ns.

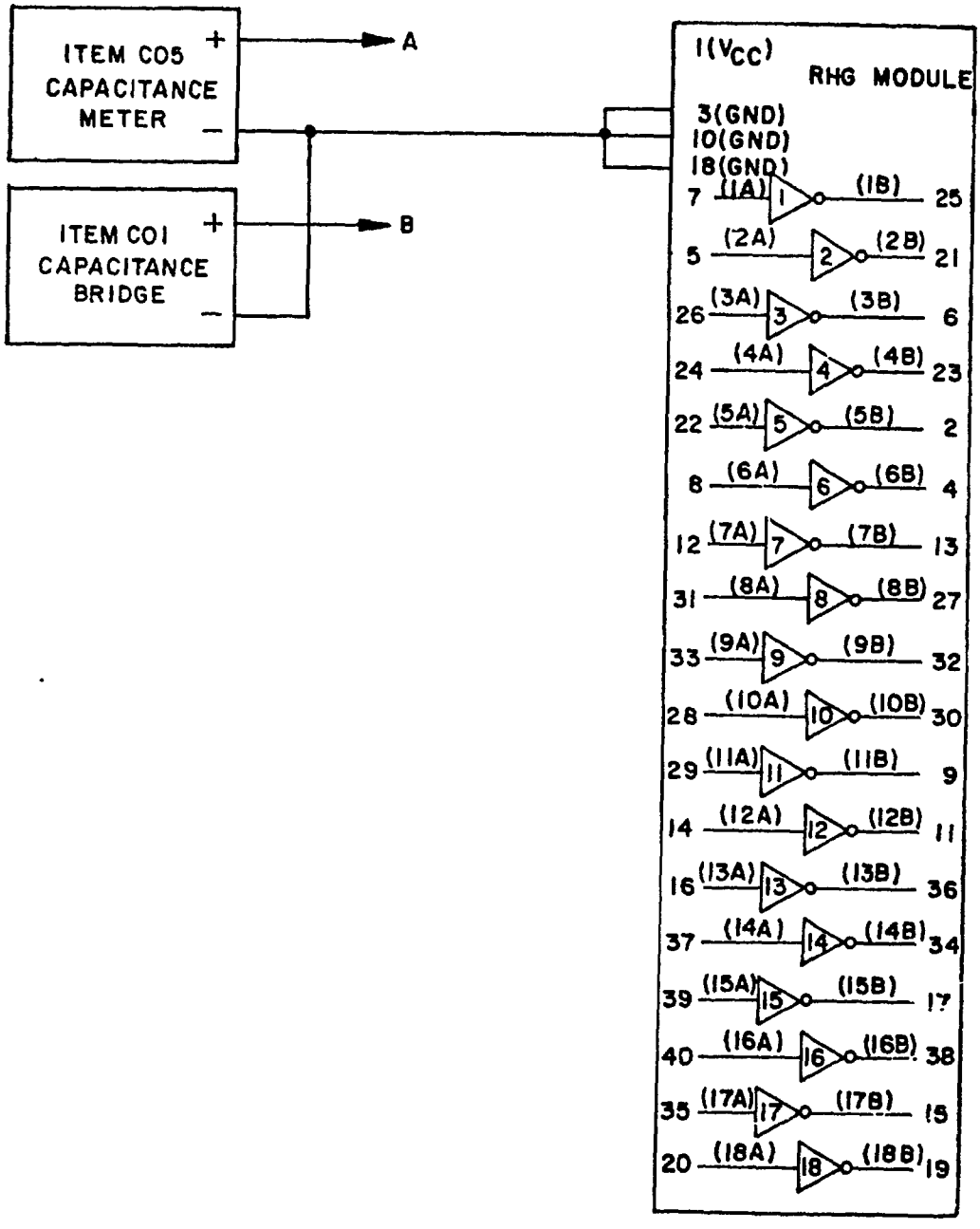
*FIGURE 22. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	1A Input capacitance	7	Open	5	15	pF	23
W2	2A "	5	"	"	"	"	"
W3	3A "	26	"	"	"	"	"
W4	4A "	24	"	"	"	"	"
W5	5A "	22	"	"	"	"	"
W6	6A "	8	"	"	"	"	"
W7	7A "	12	"	"	"	"	"
W8	8A "	31	"	"	"	"	"
W9	9A "	33	"	"	"	"	"
W10	10A "	28	"	"	"	"	"
W11	11A "	29	"	"	"	"	"
W12	12A "	14	"	"	"	"	"
W13	13A "	16	"	"	"	"	"
W14	14A "	37	"	"	"	"	"
W15	15A "	39	"	"	"	"	"
W16	16A "	40	"	"	"	"	"
W17	17A "	35	"	"	"	"	"
W18	18A "	20	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 3, and 18 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 23. Power supply filter and input capacitance test circuit for tables XIX and XX.

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3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
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